

Justin Lopato

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EDUCATION

Rice University, George R. Brown School of Engineering and Computing

Expected May 2027

Bachelor of Science in Computer Science

GPA: 3.77 / 4.0

Relevant coursework: Introduction to Modern Robotics, Computer Systems, Concurrent Program Design

WORK EXPERIENCE

Kavraki Computational Robotics Lab, Houston, TX: Research Assistant

August 2025 – Present

- Developed a trajectory visualization system for the Hello Robot Stretch using the Rerun SDK, which accelerated the debugging and analysis of motion planning algorithms
- Compared multiple implementations of SLAM mapping systems for Stretch on efficiency, usability, and data structure, to evaluate application fit for various research projects
- Implemented a geometric path planner which complied with motion constraints of default Stretch trajectory controller, eliminating the need to create a new controller to bypass constraints

Rice University, Houston, TX: Computer Systems Teaching Assistant

August 2025 – December 2025

- Mentored over 15 students in weekly lab sections and office hours, clarifying complex, low-level concepts such as bit-level data representation, x86 assembly, and the memory hierarchy
- Provided constructive feedback on students' C programming assignments, focusing on correctness, efficiency, and debugging techniques using tools like GDB and Valgrind
- Graded exams and assignments with a focus on transparency and fairness, ensuring students understood why they lost points so they could improve, rather than just pointing out errors

Rice ELITE Tech Camp, Houston, TX: Robotics Programming Instructor

May 2025 – July 2025

- Taught high school classes of 30+ students programming fundamentals in Arduino and Python classes during week-long engineering summer camps managed by the Rice Center for Engineering Leadership
- Produced lecture materials for the entire C++ curriculum, providing a solid baseline in syntax, data types, functions and arrays, as well as Arduino specific libraries like the GPIO and Serial interfaces
- Implemented custom Arduino library for stepper motor control, to aide students' use of steppers for final project

ACTIVITIES & PROJECTS

Rice Robotics Club: R&D CS Lead

February 2024 – Present

Serving as lead for the software side of a quadruped robot project, with the goal of applying machine learning to a novel mechanical design built from scratch

- Increased undergraduate member count of R&D CS team 4x through effective and consistent communication
- Revitalized team documentation to provide clear description of project goals and progress, as well as providing resources for new members to learn robotics specific terms, concepts, and tools
- Implemented prototype reinforcement learning training environment for simulation of servo-based quadruped robot
- Developed testing code for first iteration of quadruped robot leg, which enabled non-programming members to send motion commands to robot leg via a GUI
- Acquired holistic familiarity with non-CS robotics skills through robot testing, including physics, electronics, and safety
- Participated in weekly planning sessions and progress tracking, resulting in meeting all project milestones to date

Social Accountability Web App (React, Express, MongoDB)

September 2024

Created an application providing mutual academic accountability to student pairs for RiceHacks 2024 hackathon

- Achieved 70% match satisfaction based on user feedback from other hackathon participants
- Engineered vector embedding similarity system to match users with similar academic interests
- Implemented real-time chat functionality via WebSockets, enabling communication between matched users

TECHNICAL SKILLS

Programming: Python, C, C++, TypeScript, Java, Go, L^AT_EX, Docker, Data structures, Software design patterns

Software: Github, Git, Onshape, MoveIt, ROS2, OMPL, Overleaf, Postman, Notion, Slack, Ubuntu

AWARDS

Hispanic Scholarship Fund - Scholar

June 2023, June 2024, June 2025

Rice President's Honor Roll

Fall 2023, Fall 2024

Eagle Scout Award

October 2022